DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

Neural Network Lab - Assessment Report

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Overall Result

4/28 (14.3%) Grade: F

Score by Section

Section	Score	Pct
Activation Functions	0/5	0.0%
Weight Initialization	0/3	0.0%
Loss Functions	1/3	33.3%
Optimizers	0/4	0.0%
Backpropagation	0/3	0.0%
Regularization	2/4	50.0%
Batch Normalization	1/2	50.0%
Architecture Design	0/4	0.0%

Questions to Review

Q01. [Activation Functions] Correct answer: C

What is the output range of the Sigmoid activation function?

Sigmoid = 1/(1+e^-z). As z->+inf output->1, as z->-inf output->0. It never reaches 0 or 1, so the range is the open interval (0, 1).

Q02. [Activation Functions] Correct answer: C

Which activation function suffers from the 'Dying ReLU' problem?

ReLU outputs 0 for all z<=0 and its gradient is also 0 there. A neuron stuck at 0 stops receiving gradient updates and dies. Leaky ReLU fixes this with a small slope (0.01) for negative inputs.

Q03. [Activation Functions] Correct answer: B

Why is Tanh preferred over Sigmoid for hidden layers?

Tanh outputs range (-1, 1) centered around 0. Sigmoid outputs (0, 1) are always positive, causing zig-zag gradient updates. Zero-centering avoids this inefficiency.

Q04. [Activation Functions] Correct answer: C

What does the derivative of ReLU equal for z = 3.5?

ReLU derivative = 1 if z > 0, else 0. Since z=3.5 > 0, the derivative is exactly 1.

Q05. [Activation Functions] Correct answer: D

Which activation function should you use in the OUTPUT layer for a multi-class classification problem with 4 classes?

Softmax converts raw scores into a probability distribution summing to 1. Each output = P(class=k). Use Sigmoid only for binary (2-class) output.

Q06. [Weight Initialization] Correct answer: B

What problem occurs if you initialize ALL weights to zero?

With all weights=0, every neuron computes the same output and gets the same gradient. They update identically forever and symmetry is never broken.

Q07. [Weight Initialization] Correct answer: C

Which initializer is recommended when using ReLU activations?

He initialization: std = sqrt(2/fan_in). The factor 2 compensates for ReLU zeroing roughly half the neurons. Xavier uses sqrt(2/(fan_in+fan_out)), designed for Tanh/Sigmoid.

Q08. [Weight Initialization] Correct answer: B

What does 'fan_in' mean in the context of weight initialization?

fan_in = number of inputs to a layer (input dimension). fan_out = number of outputs. Xavier uses both; He uses only fan_in.

Q10. [Loss Functions] Correct answer: B

A model predicts p=0.95 for a sample with true label y=1. What is the Binary Cross-Entropy loss?

BCE = -log(0.95) = ~0.051. A confident correct prediction gives a very small loss. If p=0.05 for y=1, loss = -log(0.05) = ~3.0.

Q11. [Loss Functions] Correct answer: B

Why do we clip predictions before computing cross-entropy loss (e.g. clip to 1e-12, 1-1e-12)?

log(0) = -inf, making the loss undefined (NaN). Clipping keeps log() finite and numerically stable.

Q12. [Optimizers] Correct answer: B

What does the momentum term do in SGD with Momentum?

*Momentum: v = beta*v + lr*grad. Accelerates movement in consistent directions and dampens oscillations. Typical beta=0.9 retains 90% of previous velocity.*

Q13. [Optimizers] Correct answer: B

Adam optimizer combines which two techniques?

*Adam maintains m (first moment, like momentum) and v (second moment, like RMSProp). Update: theta -= lr * m_hat / (sqrt(v_hat) + eps).*

Q14. [Optimizers] Correct answer: C

Which optimizer typically converges fastest on small datasets?

Adam adapts the learning rate per-parameter using gradient history, making it robust to different gradient scales.

Q15. [Optimizers] Correct answer: B

What is the purpose of bias correction in Adam (dividing by 1 - beta^t)?

At t=1, m is tiny because it started at 0. Dividing by (1-beta^t) scales it back to the true gradient magnitude.

Q16. [Backpropagation] Correct answer: B

What does the chain rule tell us in backpropagation?

Chain rule: dL/dW1 = (dL/da2)(da2/dz2)*(dz2/da1)*(da1/dz1)*(dz1/dW1). Each term is a local gradient multiplied going backward.*

Q17. [Backpropagation] Correct answer: B

What is the vanishing gradient problem?

Sigmoid derivative max = 0.25. In a 10-layer net, gradient shrinks to 0.25^10 = ~0.000001 at layer 1. ReLU derivative = 1 for z>0, so gradients do not shrink.

Q18. [Backpropagation] Correct answer: B

In a gradient flow chart, what does a gradient ratio greater than 1000x between layers indicate?

1000x difference means early layers receive almost no learning signal. Fix: use ReLU, BatchNorm, or residual connections.

Q19. [Regularization] Correct answer: B

What is overfitting?

Overfitting = low train loss but high val loss. The model memorizes noise instead of learning general patterns.

Q21. [Regularization] Correct answer: B

L2 regularization adds lambda multiplied by the sum of squared weights to the loss. What effect does this have?

*L2 adds lambda*W to each gradient update, decaying weights toward 0 but never exactly 0. Also called weight decay.*

Q24. [Batch Normalization] Correct answer: B

Why does BatchNorm behave differently during training versus inference?

Training: mu and var from current mini-batch. Inference: uses stable running_mean and running_var (EMA). Needed because test batch size may be just 1.

Q25. [Architecture Design] Correct answer: B

Why does the output layer use Sigmoid and NOT ReLU for binary classification?

Binary cross-entropy expects predictions in (0,1). Sigmoid squashes any real number to (0,1). ReLU could output 5.3 which has no probabilistic meaning.

Q26. [Architecture Design] Correct answer: B

What is the purpose of using mini-batches during training with batch size 32?

Full-batch GD: exact gradient but slow. SGD (batch=1): fast but noisy. Mini-batch (32-256): balances speed and gradient quality.

Q27. [Architecture Design] Correct answer: B

Early stopping uses patience=30. What does this mean?

No improvement for 30 consecutive epochs triggers a stop and restores the weights from the best epoch seen.

Q28. [Architecture Design] Correct answer: B

Why can a single layer with no hidden layers NOT solve the XOR problem?

XOR: (0,0)->0, (0,1)->1, (1,0)->1, (1,1)->0. No straight line separates the 0s from the 1s. A hidden layer creates a non-linear transformation making classes separable.